Quiz 1

CS 102 Computer Programming I · Fall Semester 2026
Week 4 · Duration: **35 MINUTES** · Version **A**

New Uzbekistan University

FULL NAME: _____ ID: _____ GROUP: _____

Rules. No notes, phones or computers. Write answers *only* in the space given. For fill-in-the-gap questions write exactly one piece of C++ code in each numbered gap — a keyword, an operator, a type or an expression. Assume every code fragment is inside `main` with `#include <iostream>` and `using namespace std;`.

Section	Points	Score
A Multiple choice	30	
B Fill in the gaps	40	
C Trace and write	30	
Total	**100**	

Section A: Multiple Choice (30 points)

Circle **one** answer for each question.

A1. (6 points) The hexadecimal literal `0x2F` has the decimal value: A. 32 B. 47 C. 215 D. 52

A2. (6 points) What does `cout << 7 / 2 * 2.0 + 7 % 3;` print? A. 8 B. 7.5 C. 7 D. 6

A3. (6 points) What does this code print?

```
int x = 10;
if (x > 5 || x++ > 0) {
    cout << "yes ";
}
cout << x;
```

A. `yes 10` B. `yes 11` C. `11` D. `10`

A4. (6 points) Which expression does **not** evaluate to `2.5`?

A. `5 / 2.0`

B. `static_cast<double>(5) / 2`

C. `static_cast<double>(5 / 2)`

D. `5.0 / 2`

A5. (6 points) What does this print? `char c = 'A' + 2; cout << c << (c + 1);`
Hint: in the ASCII table, `'A'` *= 65.* A. `CD` B. `6768` C. `C D` D. `C68`

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Section B: Fill in the Gaps (40 points)

Each numbered gap is worth 2 points. Every program must compile and produce the output shown. Write your answers in the boxes under each program.

B1. (10 points) **Seconds to hours, minutes, seconds.** Input `3725` must print `1 h 2 min 5 s`.

```
#include <(1)_____>
using namespace std;
int main() {
    int total;
    cin (2)_____ total;
    int h = total (3)_____ 3600;
    int m = (total (4)_____ 3600) / 60;
    int s = total (5)_____ 60;
    cout << h << " h " << m << " min " << s << " s" << endl;
    return 0;
}
```

(1) _____ (2) _____ (3) _____ (4) _____ (5) _____

B2. (10 points) **Exact average, rounding and grade.** Grade `A` for an average of 90 or more, `B` for 75 or more, otherwise `C`; an average outside 0–100 is invalid. Input `85 90 78` must print `Average: 84.3333, rounded: 84, grade: B`.

```
int s1, s2, s3;
cin >> s1 >> s2 >> s3;
int sum = s1 + s2 + s3;
double avg = (6)_____(sum) / 3;    // must NOT be integer division
(7)_____ grade;
if ((8)_____) {             // outside 0..100
    cout << "Invalid" << endl;
} else {
    if (avg (9)_____ 90) {
        grade = 'A';
    } else if (avg >= 75) {
        grade = 'B';
    } else {
        grade = 'C';
    }
    cout << "Average: " << avg << ", rounded: " << (10)_____(avg + 0.5)
         << ", grade: " << grade << endl;
}
```

(6) _____ (7) _____ (8) _____ (9) _____ (10) _____

B3. (10 points) **Days in a year.** A year is a leap year if it is divisible by 4 but not by 100, *or* if it is divisible by 400. Input `2024` prints `2024 has 366 days`; `1900` prints `1900 has 365 days`; `0` prints `Invalid year`.

```
int y;
cin >> y;
if (y (11)_____ 0) {
    cout << "Invalid year" << endl;
} else {
    bool leap = (y % 4 == 0 (12)_____ y % 100 != 0) (13)_____ y % 400 == 0;
    int days = 365;
    if ((14)_____) {
        days (15)_____ 1;
    }
    cout << y << " has " << days << " days" << endl;
}
```

(11) _____ (12) _____ (13) _____ (14) _____ (15) _____

B4. (10 points) **Box category (watch the volume!).** Dimensions are up to 10^5 and mass up to 1000. A box is *bulky* if a side is at least 10^4 or its volume is at least 10^9; it is *heavy* if its mass is at least 100. Input `2000 2000 2000 50` must print `Bulky`.

```
int l, w, h, mass;
cin >> l >> w >> h >> mass;
(16) _____ volume = static_cast<long long>(l) * w * h;
bool bulky = l >= 10000 || w >= 10000 || h >= 10000 || volume >= 1000000000;
bool heavy = mass >= 100;
if (bulky (17) _____ heavy) {
    cout << "Both" << endl;
} else if ((18) _____) {
    cout << "Bulky" << endl;
} else if (heavy) {
    cout << "Heavy" << endl;
} (19) _____ {
    cout << "Neither" << endl;
}
```

(16) [ ] (17) [ ] (18) [ ] (19) [ ]

(20) Why would `int volume = l * w * h;` print `Neither` for the input above?

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Section C: Trace and Write (30 points)

C1. (15 points) Write **exactly** what this fragment prints (mind the spaces).
Hint: in the ASCII table, 'a' = 97 (and 'A' = 65).

```
int a = 7, b = 3, c = 0;
double d = a / b + a % b / 2.0;
if (a > b)
    if (c != 0 && a / c > 1)
        cout << "X";
else
    cout << "Y";
cout << " " << d;
char ch = 'a' + b;
if (ch >= 'a' && ch <= 'z')
    ch = ch - 32;
cout << " " << ch << " " << (ch > 'C') << endl;
```

C2. (15 points) **Hours after a delay.** Write a program that reads a time as hours and minutes (`hh mm`) and a delay `d` in minutes. Print the time after the delay as hours and minutes. Do **not** wrap around at 24: hours simply keep growing. Use only `int`, `/` and `%`.

```
20 0 600   → 30 h 0 min
8 45 50    → 9 h 35 min
2 30 135   → 4 h 45 min
```

END OF QUIZ